

Harsh Arya

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ABOUT ME

Full Stack Developer and AI Engineer with a robust foundation in problem-solving (300+ **Data Structures and Algorithms** challenges). Proficient in orchestrating the full development lifecycle, engineering **hyper- scalable web apps**, and architecting **advanced AI systems** utilizing **Deep Learning and Graph RAG**. Passionate about bridging the gap between modern UI/UX and cutting-edge machine learning.

PROJECTS

AlgoZen - High-Performance DSA & Competitive Programming Platform

- Architected a high-performance DSA platform using the MERN stack, supporting 100+ algorithmic challenges with a responsive Bento-grid UI built via Tailwind CSS and DaisyUI.
- Integrated Judge0 API for multi-language code execution (C++, Python, JS, etc.), implementing a
- Custom Driver Code Injection system to ensure 100% test case verification accuracy.
- Engineered a Redux-managed global activity engine that tracks real-time 365-day streaks and heatmaps, increasing user engagement by capturing granular submission data.
- Secured the application using Google OAuth 2.0 and JWT-based authorization, implementing protected routes and secure session management for a seamless "Trust Bridge" login.

Codex - Autonomous Multi-Agent AI Dev Team (LangGraph & Gemini)

- Architected an autonomous multi-agent system using LangGraph and Gemini AI, orchestrating 30+ nodes to automate the SDLC and reduce manual planning time by 70%.
- Developed a PM Agent with a Human-in-the-Loop pattern, utilizing interactive Q&A to resolve prompt ambiguities and ensure 100% requirement specification accuracy.
- Engineered a Redis-backed persistence layer for state checkpointing, ensuring zero data loss and seamless state recovery across long-running autonomous agent workflows.
- Developed a real-time Token Tracking & Cost Engine, monitoring API expenditure across multiple agents to optimize computational overhead and LLM efficiency.

GraphRAG Movie Recommendation Engine

- Built a full-stack GraphRAG movie recommendation engine using Node.js, Neo4j, Pinecone, Gemini API, and LangChain.js.
- Modeled 6+ knowledge graph entities including Movies, Actors, Directors, Genres, Themes, and Awards in Neo4j.
- Implemented 2 retrieval pipelines: graph traversal for factual queries and vector search for semantic recommendations.
- Integrated Pinecone Vector DB with Gemini embeddings for high-dimensional similarity search,
- Developed an LLM query router to classify user queries into graph or similarity workflows.

GeminiCodeReviewer - AI-Powered VS Code Extension

- Developed a custom VS Code extension leveraging Generative AI (Gemini API) to serve as an intelligent, autonomous code review and repair agent.
- Streamlined developer workflows by implementing real-time error highlighting (Diagnostics), a rightclick "Fix Selection" command, and 1-click resolutions via CodeActionProvider.

Skills

Languages: JavaScript (ES6+), TypeScript, Python, HTML5, CSS3, Node.js, C++

AI, Data & Databases: Deep Learning, Graph RAG, LangGraph, LangChain, Multi-Agent Systems, Vector Databases (Pinecone), Graph Databases (Neo4j), Generative AI (Gemini API).

Libraries & Frameworks: React, Redux Toolkit, React Router, Tailwind CSS, Daisy UI, Web Sockets

Tools & Platforms: Docker, Redis, Git, GitHub, REST APIs, Netlify, Vercel, VS Code Extension API, MongoDB, postgresql JWT(JSON Web Token).

Concepts: Agentic Workflows, Human-in-the-Loop AI, System Architecture, State Management, Full-Stack Development, Problem Solving.

Education

Bachelor of Technology in Computer Science and Engineering

2023 - 2027

Senior Secondary (XII), Uttar Pradesh Board

2020 - 2021

B.D. Inter College, Sambhal, Uttar Pradesh